R Reference Card for Data Mining

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- See the latest version at `http://www.RDataMining.com`
- The package names are in parentheses.
- Recommended packages and functions are shown in **bold**.
- Click a package in this PDF file to find it on CRAN.

Association Rules and Sequential Patterns

Functions

`apriori()` mine associations with APRIORI algorithm – a level-wise, breadth-first algorithm which counts transactions to find frequent itemsets (*arules*)

`eclat()` mine frequent itemsets with the Eclat algorithm, which employs equivalence classes, depth-first search and set intersection instead of counting (*arules*)

`cspade()` mine frequent sequential patterns with the cSPADE algorithm (*arulesSequences*)

`seqefsub()` search for frequent subsequences (*TraMineR*)

Packages

arules mine frequent itemsets, maximal frequent itemsets, closed frequent itemsets and association rules. It includes two algorithms, Apriori and Eclat.
arulesViz visualizing association rules
arulesSequences add-on for *arules* to handle and mine frequent sequences
TraMineR mining, describing and visualizing sequences of states or events
arulesCBA, *arc*, *rCBA* classification based on association rules

Classification & Prediction

Decision Trees

`ctree()` conditional inference trees, recursive partitioning for continuous, censored, ordered, nominal and multivariate response variables in a conditional inference framework (*party*)

`rpart()` recursive partitioning and regression trees (*rpart*)

`mob()` model-based recursive partitioning, yielding a tree with fitted models associated with each terminal node (*party*)

Random Forest

`cforest()` random forest and bagging ensemble (*party*)

`randomForest()` random forest (*randomForest*)

`importance()` variable importance (*randomForest*)

`varimp()` variable importance (*party*)

Neural Networks

`nnet()` fit single-hidden-layer neural network (*nnet*)

`mlp(), dlvq(), rbf(), rbfDDA(), elman(), jordan(), som(), art1(), art2(), artmap(), assoz()` various types of neural networks (*RSNNS*)

`neuralnet` training of neural networks (*neuralnet*)

Support Vector Machine (SVM)

`svm()` train a support vector machine for regression, classification or density-estimation (*e1071*)

`ksvm()` support vector machines (*kernlab*)

Bayes Classifiers

`naiveBayes()` naive Bayes classifier (*e1071*)

Performance Evaluation

`performance()` provide various measures for evaluating performance of prediction and classification models (*ROCR*)

`PRcurve()` precision-recall curves (*DMwR*)

`CRchart()` cumulative recall charts (*DMwR*)

`roc()` build a ROC curve (*pROC*)

`auc()` compute the area under the ROC curve (*pROC*)

`ROC()` draw a ROC curve (*DiagnosisMed*)

Packages

party recursive partitioning
rpart recursive partitioning and regression trees
randomForest classification and regression based on a forest of trees using random inputs
ParallelForest random forest classification with parallel computing
ROCR visualize the performance of scoring classifiers
caret classification and regression models
r1071 functions for latent class analysis, short time Fourier transform, fuzzy clustering, support vector machines, shortest path computation, bagged clustering, naive Bayes classifier, ...
rpartOrdinal ordinal classification trees, deriving a classification tree when the response to be predicted is ordinal
rpart.plot plots *rpart* models
pROC display and analyze ROC curves
nnet feed-forward neural networks and multinomial log-linear models
RSNNS neural networks in R using the Stuttgart Neural Network Simulator (SNNS)
neuralnet training of neural networks using backpropagation, resilient backpropagation with or without weight backtracking

Regression

Functions

`lm()` linear regression

`glm()` generalized linear regression

`gbm()` generalized boosted regression models (*gbm*)

`predict()` predict with models

`residuals()` residuals, the difference between observed values and fitted values

`nls()` non-linear regression

`gls()` fit a linear model using generalized least squares (*nlme*)

`gnls()` fit a nonlinear model using generalized least squares (*nlme*)

Packages

nlme linear and nonlinear mixed effects models
gbm generalized boosted regression models

Clustering

Partitioning based Clustering

partition the data into k groups first and then try to improve the quality of clustering by moving objects from one group to another

`kmeans()` perform k-means clustering on a data matrix

`kmeansruns()` call `kmeans` for the k-means clustering method and includes estimation of the number of clusters and finding an optimal solution from several starting points (*fpc*)

`pam()` the Partitioning Around Medoids (PAM) clustering method (*cluster*)

`pamk()` the Partitioning Around Medoids (PAM) clustering method with estimation of number of clusters (*fpc*)

`kmeansCBI()` interface function for `kmeans` (*fpc*)

`cluster.optimal()` search for the optimal k-clustering of the dataset (*bayesclust*)

`clara()` Clustering Large Applications (*cluster*)

`fanny(x,k,...)` compute a fuzzy clustering of the data into `k` clusters (*cluster*)

`kcca()` k-centroids clustering (*flexclust*)

`ccfkms()` clustering with Conjugate Convex Functions (*cba*)

`apcluster()` affinity propagation clustering for a given similarity matrix (*apcluster*)

`apclusterK()` affinity propagation clustering to get K clusters (*apcluster*)

`cclust()` Convex Clustering, incl. k-means and two other clustering algorithms (*cclust*)

`KMeansSparseCluster()` sparse k-means clustering (*sparcl*)

`tclust(x,k,alpha,...)` trimmed k-means with which a proportion `alpha` of observations may be trimmed (*tclust*)

Hierarchical Clustering

a hierarchical decomposition of data in either bottom-up (agglomerative) or top-down (divisive) way

`hclust()` hierarchical cluster analysis on a set of dissimilarities

`birch()` the BIRCH algorithm that clusters very large data with a CF-tree (*birch*)

`pvclust()` hierarchical clustering with p-values via multi-scale bootstrap resampling (*pvclust*)

`agnes()` agglomerative hierarchical clustering (*cluster*)

`diana()` divisive hierarchical clustering (*cluster*)

`mona()` divisive hierarchical clustering of a dataset with binary variables only (*cluster*)

`rockCluster()` cluster a data matrix using the Rock algorithm (*cba*)

`proximus()` cluster the rows of a logical matrix using the Proximus algorithm (*cba*)

`isopam()` Isopam clustering algorithm (*isopam*)

`flashClust()` optimal hierarchical clustering (*flashClust*)

`fastcluster()` fast hierarchical clustering (*fastcluster*)

`cutreeDynamic(), cutreeHybrid()` detection of clusters in hierarchical clustering dendrograms (*dynamicTreeCut*)

`HierarchicalSparseCluster()` hierarchical sparse clustering (*sparcl*)

Model based Clustering

`Mclust()` model-based clustering (*mclust*)

`HDDC()` a model-based method for high dimensional data clustering (*HDclassif*)

`fixmahal()` Mahalanobis Fixed Point Clustering (*fpc*)

`fixreg()` Regression Fixed Point Clustering (*fpc*)

`mergenormals()` clustering by merging Gaussian mixture components (*fpc*)

Density based Clustering

generate clusters by connecting dense regions

`dbscan(data,eps,MinPts,...)` generate a density based clustering of arbitrary shapes, with neighborhood radius set as `eps` and density threshold as `MinPts` (*fpc*)

`pdfCluster()` clustering via kernel density estimation (*pdfCluster*)

Other Clustering Techniques

`mixer()` random graph clustering (*mixer*)
`nncluster()` fast clustering with restarted minimum spanning tree (*nnclust*)
`orclus()` ORCLUS subspace clustering (*orclus*)

Plotting Clustering Solutions

`plotcluster()` visualisation of a clustering or grouping in data (*fpc*)
`bannerplot()` a horizontal barplot visualizing a hierarchical clustering (*cluster*)

Cluster Validation

`silhouette()` compute or extract silhouette information (*cluster*)
`cluster.stats()` compute several cluster validity statistics from a clustering and a dissimilarity matrix (*fpc*)
`clValid()` calculate validation measures for a given set of clustering algorithms and number of clusters (*clValid*)
`clustIndex()` calculate the values of several clustering indexes, which can be independently used to determine the number of clusters existing in a data set (*cclust*)
`NbClust()` provide 30 indices for cluster validation and determining the number of clusters (*NbClust*)

Packages

cluster cluster analysis
fpc various methods for clustering and cluster validation
mclust model-based clustering and normal mixture modeling
birch clustering very large datasets using the BIRCH algorithm
pvclust hierarchical clustering with p-values
apcluster Affinity Propagation Clustering
cclust Convex Clustering methods, including k-means algorithm, On-line Update algorithm and Neural Gas algorithm and calculation of indexes for finding the number of clusters in a data set
cba Clustering for Business Analytics, including clustering techniques such as Proximus and Rock
bclust Bayesian clustering using spike-and-slab hierarchical model, suitable for clustering high-dimensional data
biclust algorithms to find bi-clusters in two-dimensional data
clue cluster ensembles
clues clustering method based on local shrinking
clValid validation of clustering results
clv cluster validation techniques, contains popular internal and external cluster validation methods for outputs produced by package *cluster*
bayesclust tests/searches for significant clusters in genetic data
clustsig significant cluster analysis, tests to see which (if any) clusters are statistically different
clusterSim search for optimal clustering procedure for a data set
clusterGeneration random cluster generation
gcExplorer graphical cluster explorer
hybridHclust hybrid hierarchical clustering via mutual clusters
Modalclust hierarchical modal Clustering
iCluster integrative clustering of multiple genomic data types
EMCC evolutionary Monte Carlo (EMC) methods for clustering
rEMM extensible Markov Model (EMM) for data stream clustering

Outlier Detection

Functions

`boxplot.stats()$out` list data points lying beyond the extremes of the whiskers
`lofactor()` calculate local outlier factors using the LOF algorithm (*DMwR* or *dprep*)
`lof()` a parallel implementation of the LOF algorithm (*Rlof*)

Packages

Rlof a parallel implementation of the LOF algorithm
extremevalues detect extreme values in one-dimensional data
mvoutlier multivariate outlier detection based on robust methods
outliers some tests commonly used for identifying outliers

Time Series Analysis

Construction & Plot

`ts()` create time-series objects
`plot.ts()` plot time-series objects
`smoothts()` time series smoothing (*ast*)
`sfilter()` remove seasonal fluctuation using moving average (*ast*)

Decomposition

`decomp()` time series decomposition by square-root filter (*timsac*)
`decompose()` classical seasonal decomposition by moving averages
`stl()` seasonal decomposition of time series by `loess`
`tsr()` time series decomposition (*ast*)
`ardec()` time series autoregressive decomposition (*ArDec*)

Forecasting

`arima()` fit an ARIMA model to a univariate time series
`predict.Arima()` forecast from models fitted by `arima`
`auto.arima()` fit best ARIMA model to univariate time series (*forecast*)
`forecast.stl(), forecast.ets(), forecast.Arima()` forecast time series using `stl`, `ets` and `arima` models (*forecast*)

Correlation and Covariance

`acf()` autocovariance or autocorrelation of a time series
`ccf()` cross-correlation or cross-covariance of two univariate series

Packages

forecast displaying and analysing univariate time series forecasts
hts analysing and forecasting hierarchical and grouped time series
TSclust time series clustering utilities
dtw Dynamic Time Warping (DTW)
timsac time series analysis and control program
ast time series analysis
ArDec time series autoregressive-based decomposition
dse tools for multivariate, linear, time-invariant, time series models

Text Mining

Importing Text

`readPDF()` extract text and metadata from a PDF document (*tm*)

Text Cleaning and Preparation

`Corpus()` build a corpus, which is a collection of text documents (*tm*)
`tm_map()` transform text documents, e.g., stemming, stopword removal (*tm*)
`tm_filter()` filtering out documents (*tm*)
`TermDocumentMatrix(), DocumentTermMatrix()` construct a term-document matrix or a document-term matrix (*tm*)
`Dictionary()` construct a dictionary from a character vector or a term-document matrix (*tm*)
`stemDocument()` stem words in a text document (*tm*)
`stemCompletion()` complete stemmed words (*tm*)
`SnowballStemmer()` Snowball word stemmers (*Snowball*)
`stopwords(language)` return stopwords in different languages (*tm*)
`removeNumbers(), removePunctuation(), removeWords()` remove numbers, punctuation marks, or a set of words from a text document (*tm*)
`removeSparseTerms()` remove sparse terms from a term-document matrix (*tm*)

Frequent Terms and Association

`findAssocs()` find associations in a term-document matrix (*tm*)
`findFreqTerms()` find frequent terms in a term-document matrix (*tm*)
`termFreq()` generate a term frequency vector from a text document (*tm*)

Topic Modelling

`LDA()` fit a LDA (latent Dirichlet allocation) model (*topicmodels*)
`CTM()` fit a CTM (correlated topics model) model (*topicmodels*)
`terms()` extract the most likely terms for each topic (*topicmodels*)
`topics()` extract the most likely topics for each document (*topicmodels*)

Sentiment Analysis

`calculate_score()` calculate the score of sentences (*RSentiment*)
`calculate_sentiment()` calculate the sentiment of sentences (*RSentiment*)
`sentiment()` approximate the sentiment (polarity) of text by sentence (*sentimentr*)
`analyzeSentiment()` performs sentiment analysis of given object (vector of strings, document-term matrix, corpus) (*SentimentAnalysis*)
`sentiment()` Twitter sentiment text analysis (*sentiment140*)
`polarity()` polarity score (*qdap*)

Text Categorization

`textcat()` n-gram based text categorization (*textcat*)

Text Visualizatoin

`wordcloud()` plot a word cloud (*wordcloud*)
`comparison.cloud()` plot a cloud comparing the frequencies of words across documents (*wordcloud*)
`commonality.cloud()` plot a cloud of words shared across documents (*wordcloud*)

Packages

tm a framework for text mining applications
text2vec Fast and memory-friendly tools for text vectorization, topic modeling (LDA, LSA), word embeddings (GloVe), similarities
topicmodels fit topic models with LDA and CTM
wordcloud various word clouds
lda fit topic models with LDA
wordnet an interface to the WordNet
RTextTools automatic text classification via supervised learning
qdap transcript analysis, text mining and natural language processing
sentimentr calculate text polarity sentiment
RSentiment analyse sentiment of English sentences
SentimentAnalysis dictionary-based sentiment analysis
sentiment140 Twitter sentiment text analysis

tm.plugin.dc a plug-in for package *tm* to support distributed text mining
tm.plugin.mail a plug-in for package *tm* to handle mail
textir a suite of tools for inference about text documents and associated sentiment
tau utilities for text analysis
textcat n-gram based text categorization
Rwordseg Chinese word segmentation using Ansj

Social Network Analysis and Graph Mining

Functions

`graph()`, `graph.edgelist()`, `graph.adjacency()`, `graph.incidence()` create graph objects respectively from edges, an edge list, an adjacency matrix and an incidence matrix (*igraph*)
`plot()`, `tkplot()`, `rglplot()` static, interactive and 3D plotting of graphs (*igraph*)
`gplot()`, `gplot3d()` plot graphs (*sna*)
`vcount()`, `ecount()` number of vertices/edges (*igraph*)
`V()`, `E()` vertex/edge sequence of igraph (*igraph*)
`is.directed()` whether the graph is directed (*igraph*)
`are.connected()` check whether two nodes are connected (*igraph*)
`degree()`, `betweenness()`, `closeness()`, `transitivity()`, `evcent()` various centrality measures (*igraph*, *sna*)
`edge_density()` density of a graph (*igraph*)
`add.edges()`, `add.vertices()`, `delete.edges()`, `delete.vertices()` add and delete edges and vertices (*igraph*)
`neighborhood()` neighborhood of graph vertices (*igraph*, *sna*)
`get.adjlist()` adjacency lists for edges or vertices (*igraph*)
`nei()`, `adj()`, `from()`, `to()` vertex/edge sequence indexing (*igraph*)
`cliques()`, `largest.cliques()`, `maximal.cliques()`, `clique.number()` find cliques, ie. complete subgraphs (*igraph*)
`clusters()`, `no.clusters()` maximal connected components of a graph and the number of them (*igraph*)
`fastgreedy.community()`, `spinglass.community()` community detection (*igraph*)
`cohesive.blocks()` calculate cohesive blocks (*igraph*)
`induced.subgraph()` create a subgraph of a graph (*igraph*)
`mst()` minimum spanning tree (*igraph*)
`components()` calculate the maximal connected components (*igraph*)
`shortest_paths()` the shortest paths between vertices (*igraph*)
`%->%`, `%<-%`, `%--%` edge sequence indexing (*igraph*)
`get.edgelist()` return an edge list in a two-column matrix (*igraph*)
`read.graph()`, `write.graph()` read and writ graphs from and to files of various formats (*igraph*)

Packages

igraph network analysis and visualization
sna social network analysis
d3Network, ***networkD3*** creating D3 JavaScript network, tree, dendrogram, and Sankey graphs from R
RNeo4j interact with a Neo4j database through R
statnet a set of tools for the representation, visualization, analysis and simulation of network data
egonet ego-centric measures in social network analysis
snort social network-analysis on relational tables
network tools to create and modify network objects
bipartite visualising bipartite networks and calculating some (ecological) indices
blockmodeling generalized and classical blockmodeling of valued networks
diagram visualising simple graphs (networks), plotting flow diagrams
NetCluster clustering for networks
NetData network data for McFarland's SNA R labs
NetIndices estimating network indices, including trophic structure of foodwebs in R
NetworkAnalysis statistical inference on populations of weighted or unweighted networks
tnet analysis of weighted, two-mode, and longitudinal networks

Deep Learning

Packages

kerasR R interface to the Keras Deep Learning Library
keras R interface to Keras, developed by RStudio

Recommended Readings

keras - Deep Learning in R
Keras: The Python Deep Learning library

Spatial Data Analysis

Functions

`geocode()` geocodes a location using Google Maps (*ggmap*)
`plotGoogleMaps()` create a plot of spatial data on Google Maps (*plotGoogleMaps*)
`qmap()` quick map plot (*ggmap*)
`get_map()` queries the Google Maps, OpenStreetMap, or Stamen Maps server for a map at a certain location (*ggmap*)
`gvisGeoChart()`, `gvisGeoMap()`, `gvisIntensityMap()`, `gvisMap()` Google geo charts and maps (*googleVis*)
`GetMap()` download a static map from the Google server (*RgoogleMaps*)
`ColorMap()` plot levels of a variable in a colour-coded map (*RgoogleMaps*)
`PlotOnStaticMap()` overlay plot on background image of map tile (*RgoogleMaps*)
`TextOnStaticMap()` plot text on map (*RgoogleMaps*)

Packages

plotGoogleMaps plot spatial data as HTML map mushup over Google Maps
RgoogleMaps overlay on Google map tiles in R
ggmap Spatial visualization with Google Maps and OpenStreetMap
plotKML visualization of spatial and spatio-temporal objects in Google Earth
SGCS Spatial Graph based Clustering Summaries for spatial point patterns
spdep spatial dependence: weighting schemes, statistics and models

Statistics

Summarization

`summary()` summarize data
`describe()` concise statistical description of data (*Hmisc*)
`boxplot.stats()` box plot statistics

Analysis of Variance

`aov()` fit an analysis of variance model
`anova()` compute analysis of variance (or deviance) tables for one or more fitted model objects

Statistical Tests

`chisq.test()` chi-squared contingency table tests and goodness-of-fit tests
`ks.test()` Kolmogorov-Smirnov tests
`t.test()` student's t-test
`prop.test()` test of equal or given proportions
`binom.test()` exact binomial test

Mixed Effects Models

`lme()` fit a linear mixed-effects model (*nlme*)
`nlme()` fit a nonlinear mixed-effects model (*nlme*)

Principal Components and Factor Analysis

`princomp()` principal components analysis
`prcomp()` principal components analysis

Other Functions

`var()`, `cov()`, `cor()` variance, covariance, and correlation
`density()` compute kernel density estimates
`cmdscale()` Multidimensional Scaling (MDS)

Packages

nlme linear and nonlinear mixed effects models

Graphics

Functions

`plot()` generic function for plotting
`barplot()`, `pie()`, `hist()` bar chart, pie chart and histogram
`boxplot()` box-and-whisker plot
`stripchart()` one dimensional scatter plot
`dotchart()` Cleveland dot plot
`qqnorm()`, `qqplot()`, `qqline()` QQ (quantile-quantile) plot
`coplot()` conditioning plot
`splom()` conditional scatter plot matrices (*lattice*)
`pairs()` a matrix of scatterplots
`cpairs()` enhanced scatterplot matrix (*gclus*)
`parcoord()` parallel coordinate plot (*MASS*)
`cparcoord()` enhanced parallel coordinate plot (*gclus*)
`parallelplot()` parallel coordinates plot (*lattice*)
`densityplot()` kernel density plot (*lattice*)
`contour()`, `filled.contour()` contour plot
`levelplot()`, `contourplot()` level plots and contour plots (*lattice*)
`smoothScatter()` scatterplots with smoothed densities color representation; capable of visualizing large datasets
`sunflowerplot()` a sunflower scatter plot
`assocplot()` association plot
`mosaicplot()` mosaic plot
`matplot()` plot the columns of one matrix against the columns of another
`fourfoldplot()` a fourfold display of a $2 \times 2 \times k$ contingency table
`persp()` perspective plots of surfaces over the x?y plane
`cloud()`, `wireframe()` 3d scatter plots and surfaces (*lattice*)
`interaction.plot()` two-way interaction plot
`iplot()`, `ihist()`, `ibar()`, `ipcp()` interactive scatter plot, histogram, bar plot, and parallel coordinates plot (*iplots*)
`pdf()`, `postscript()`, `win.metafile()`, `jpeg()`, `bmp()`, `png()`, `tiff()` save graphs into files of various formats
`gvisAnnotatedTimeLine()`, `gvisAreaChart()`, `gvisBarChart()`, `gvisBubbleChart()`,

gvisCandlestickChart(), gvisColumnChart(), gvisComboChart(), gvisGauge(), gvisGeoChart(), gvisGeoMap(), gvisIntensityMap(), gvisLineChart(), gvisMap(), gvisMerge(), gvisMotionChart(), gvisOrgChart(), gvisPieChart(), gvisScatterChart(), gvisSteppedAreaChart(), gvisTable(), gvisTreeMap() various interactive charts produced with the Google Visualisation API (*googleVis*)

gvisMerge() merge two *googleVis* charts into one (*googleVis*)

Packages

ggplot2 an implementation of the Grammar of Graphics
ggvis interactive grammar of graphics
googleVis an interface between R and the Google Visualisation API to create interactive charts
d3Network, ***networkD3*** creating D3 JavaScript network, tree, dendrogram, and Sankey graphs from R
rCharts interactive javascript visualizations from R
lattice a powerful high-level data visualization system, with an emphasis on multivariate data
vcd visualizing categorical data
iplots interactive graphics

Data Manipulation

Functions

`transform()` transform a data frame
scale() scaling and centering of matrix-like objects
t() matrix transpose
`aperm()` array transpose
sample() sampling
table(), tabulate(), xtabs() cross tabulation
stack(), unstack() stacking vectors
split(), unsplit() divide data into groups and reassemble
reshape() reshape a data frame between "wide" and "long" format
merge() merge two data frames; similar to database `join` operations
aggregate() compute summary statistics of data subsets
by() apply a function to a data frame split by factors
melt(), cast() melt and then cast data into the reshaped or aggregated form you want (*reshape*)
`complete.cases()` find complete cases, i.e., cases without missing values
na.fail, na.omit, na.exclude, na.pass handle missing values

Packages

dplyr a fast, consistent tool for working with data frame like objects
reshape flexibly restructure and aggregate data using melt and cast
reshape2 flexibly reshape data: a reboot of the *reshape* package
tidyr easily tidy data with spread and gather functions; an evolution of reshape2
data.table extension of data.frame for fast indexing, ordered joins, assignment, and grouping and list columns
gdata various tools for data manipulation
lubridate functions to work with data and time
stringr string operations

Data Access

Functions

save(), load() save and load R data objects
read.csv(), write.csv() import from and export to .CSV files
read.table(), write.table(), scan(), write() read and write data
read_xlsx read Excel files (*readxl*)
read.xlsx(), write.xlsx() read and write Excel files (*xlsx*)
read.fwf() read fixed width format files
`write.matrix()` write a matrix or data frame (*MASS*)
readLines(), writeLines() read/write text lines from/to a connection, such as a text file
sqlQuery() submit an SQL query to an ODBC database (*RODBC*)
sqlFetch() read a table from an ODBC database (*RODBC*)
sqlSave(), sqlUpdate() write or update a table in an ODBC database (*RODBC*)
`sqlColumns()` enquire about the column structure of tables (*RODBC*)
`sqlTables()` list tables on an ODBC connection (*RODBC*)
odbcConnect(), odbcClose(), odbcCloseAll() open/close connections to ODBC databases (*RODBC*)
`dbSendQuery` execute an SQL statement on a given database connection (*DBI*)
dbConnect(), dbDisconnect() create/close a connection to a DBMS (*DBI*)

Packages

RODBC ODBC database access
foreign read and write data in other formats, such as Minitab, S, SAS, SPSS, Stata, Systat, ...
sqldf perform SQL selects on R data frames
DBI a database interface (DBI) between R and relational DBMS
RMySQL interface to the MySQL database
RJDBC access to databases through the JDBC interface
RSQLite SQLite interface for R
ROracle Oracle database interface (DBI) driver
RpgSQL DBI/RJDBC interface to PostgreSQL database
RODM interface to Oracle Data Mining
readxl, ***openxlsx***, ***xlsx*** read and write Excel files
xlsReadWrite read and write Excel files
WriteXLS create Excel 2003 (XLS) files from data frames
SPARQL Use SPARQL to pose SELECT or UPDATE queries to an end-point

Web Data Access

Functions

download.file() download a file from the Internet
xmlParse(), htmlParse() parse an XML or HTML file (*XML*)
userTimeline(), homeTimeline(), mentions(), retweetsOfMe() retrieve various timelines within the Twitter universe (*twitteR*)
searchTwitter() a search of Twitter based on a supplied search string (*twitteR*)
getUser(), lookupUsers() get information of Twitter users (*twitteR*)
getFollowers(), getFollowerIDs(), getFriends(), getFriendIDs() get a list of followers/friends or their IDs of a Twitter user (*twitteR*)
twListToDF() convert *twitteR* lists to data frames (*twitteR*)

Packages

twitteR an interface to the Twitter web API
RCurl general network (HTTP/FTP/...) client interface for R
XML reading and creating XML and HTML documents
httr tools for working with URLs and HTTP; a simplified wrapper built on top of *RCurl*

MapReduce, Hadoop and Spark

Functions

`mapreduce()` define and execute a MapReduce job (***rmr2***)
`keyval()` create a key-value object (***rmr2***)
`from.dfs()`, `to.dfs()` read/write R objects from/to file system (***rmr2***)
`hb.get()`, `hb.scan()`, `hb.get.data.frame()` read HBase tables (***rhbase***)
`hb.insert()`, `hb.insert.data.frame()` write to HBase tables (***rhbase***)
`hb.delete()` delete from HBase tables (***rhbase***)

Packages

rmr2 perform data analysis with R via MapReduce on a Hadoop cluster
rhdfs connect to the Hadoop Distributed File System (HDFS)
rhbase connect to the NoSQL HBase database
Rhipe R and Hadoop Integrated Processing Environment
SparkR a light-weight frontend to use Apache Spark from R
RHive distributed computing via HIVE query
Segue Parallel R in the cloud using Amazon's Elastic Map Reduce (EMR) engine
HadoopStreaming Utilities for using R scripts in Hadoop streaming
hive distributed computing via the MapReduce paradigm
rHadoopClient Hadoop client interface for R

Large Data

Functions

`as.ffdf()` coerce a dataframe to an `ffdf` (*ff*)
`read.table.ffdf()`, `read.csv.ffdf()` read data from a flat file to an `ffdf` object (*ff*)
`write.table.ffdf()`, `write.csv.ffdf()` write an `ffdf` object to a flat file (*ff*)
`ffdfappend()` append a dataframe or an `ffdf` to an existing `ffdf` (*ff*)
`big.matrix()` create a standard `big.matrix`, which is constrained to available RAM (*bigmemory*)
`read.big.matrix()` create a `big.matrix` by reading from an ASCII file (*bigmemory*)
`write.big.matrix()` write a `big.matrix` to a file (*bigmemory*)
`filebacked.big.matrix()` create a file-backed `big.matrix`, which may exceed available RAM by using hard drive space (*bigmemory*)
`mwhich()` expanded "which"-like functionality (*bigmemory*)

Packages

ff memory-efficient storage of large data on disk and fast access functions
ffbase basic statistical functions for package *ff*
filehash a simple key-value database for handling large data
g.data create and maintain delayed-data packages
BufferedMatrix a matrix data storage object held in temporary files
biglm regression for data too large to fit in memory
bigmemory manage massive matrices with shared memory and memory-mapped files
biganalytics extend the *bigmemory* package with various analytics

bigtabulate table-, tapply-, and split-like functionality for matrix and `big.matrix` objects

Parallel Computing

Functions

`sfInit()`, `sfStop()` initialize and stop the cluster (*snowfall*)
`sfLapply()`, `sfSapply()`, `sfApply()` parallel versions of `lapply()`, `sapply()`, `apply()` (*snowfall*)
`foreach(...)` `%dopar%` looping in parallel (*foreach*)
`registerDoSEQ()`, `registerDoSNOW()`, `registerDoMC()` register respectively the sequential, SNOW and multicore parallel backend with the *foreach* package (*foreach*, *doSNOW*, *doMC*)

Packages

parallel support for parallel computation
snowfall usability wrapper around *snow* for easier development of parallel R programs
snow simple parallel computing in R
multicore parallel processing of R code on machines with multiple cores or CPUs
snowFT extension of *snow* supporting fault tolerant and reproducible applications, and easy-to-use parallel programming
Rmpi interface (Wrapper) to MPI (Message-Passing Interface)
rpvm R interface to PVM (Parallel Virtual Machine)
nws provide coordination and parallel execution facilities
foreach foreach looping construct for R
doMC foreach parallel adaptor for the *multicore* package
doSNOW foreach parallel adaptor for the *snow* package
doMPI foreach parallel adaptor for the *Rmpi* package
doParallel foreach parallel adaptor for the *multicore* package
doRNG generic reproducible parallel backend for `foreach` Loops
GridR execute functions on remote hosts, clusters or grids
fork R functions for handling multiple processes

Interface to Weka

Package ***RWeka*** is an R interface to Weka, and enables to use the following Weka functions in R.
Association rules:
`Apriori()`, `Tertius()`
Regression and classification:
`LinearRegression()`, `Logistic()`, `SMO()`
Lazy classifiers:
`IBk()`, `LBR()`
Meta classifiers:
`AdaBoostM1()`, `Bagging()`, `LogitBoost()`, `MultiBoostAB()`, `Stacking()`, `CostSensitiveClassifier()`
Rule classifiers:
`JRip()`, `M5Rules()`, `OneR()`, `PART()`
Regression and classification trees:
`J48()`, `LMT()`, `M5P()`, `DecisionStump()`
Clustering:
`Cobweb()`, `FarthestFirst()`, `SimpleKMeans()`, `XMeans()`, `DBScan()`
Filters:
`Normalize()`, `Discretize()`
Word stemmers:
`IteratedLovinsStemmer()`, `LovinsStemmer()`
Tokenizers:
`AlphabeticTokenizer()`, `NGramTokenizer()`, `WordTokenizer()`

Interface to Other Programming Languages

Functions

`.jcall()` call a Java method (*rJava*)
`.jnew()` create a new Java object (*rJava*)
`.jinit()` initialize the Java Virtual Machine (JVM) (*rJava*)
`.jaddClassPath()` adds directories or JAR files to the class path (*rJava*)

Packages

rJava low-level R to Java interface
rPython call Python from R
reticulate R Interface to Python

Generating Documents and Reports

Functions

`Sweave()` mixing text and R/S code for automatic report generation
`xtable()` export tables to LaTeX or HTML (*xtable*)

Packages

knitr a general-purpose package for dynamic report generation in R
xtable export tables to LaTeX or HTML
R2HTML making HTML reports
R2PPT generating Microsoft PowerPoint presentations

Building GUIs and Web Applications

shiny web application framework for R
svDialogs dialog boxes
gWidgets a toolkit-independent API for building interactive GUIs

R Editors/GUIs

RStudio a free integrated development environment (IDE) for R
Tinn-R a free GUI for R language and environment
rattle graphical user interface for data mining in R
Rpad workbook-style, web-based interface to R
RPMG graphical user interface (GUI) for interactive R analysis sessions
Red-R An open source visual programming GUI interface for R
R AnalyticFlow a software which enables data analysis by drawing analysis flowcharts
latticist a graphical user interface for exploratory visualisation

R Reference Cards

R Reference Card, by Tom Short
R Reference Card, by Jonathan Baron
R Functions for Regression Analysis, by Vito Ricci
R Functions for Time Series Analysis, by Vito Ricci

RDataMining Books

R and Data Mining: Examples and Case Studies
introduces into using R for data mining with examples and case studies.
`http://www.rdatamining.com/docs/RDataMining-book.pdf`
Data Mining Applications with R
presents 15 real-world applications on data mining with R.
`http://www.rdatamining.com/books/dmar`

RDataMining Website, Group & Twitter

RDataMining Website
`http://www.rdatamining.com`
RDataMining Group on LinkedIn
`http://group.rdatamining.com` or
`https://www.linkedin.com/groups/4066593`
RDataMining on Twitter
@RDataMining

Comments & Feedback

If you have questions on using R for data mining, please post them to the RDataMining Group on LinkedIn at `http://group.rdatamining.com`.

If you have any comments on this reference card, or would like to suggest any relevant R packages or functions, please feel free to email me <yanchang@rdatamining.com>. Thanks.